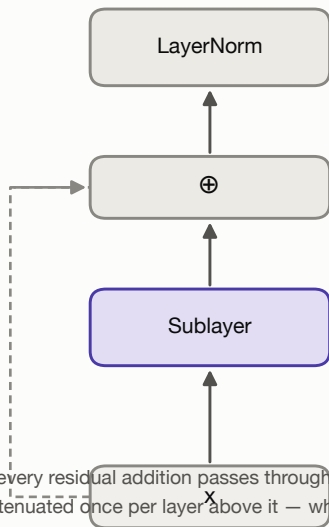


Where to put the layer normalisation

Post-LN (original paper)

$$x \leftarrow \text{LayerNorm}(x + \text{Sublayer}(x))$$

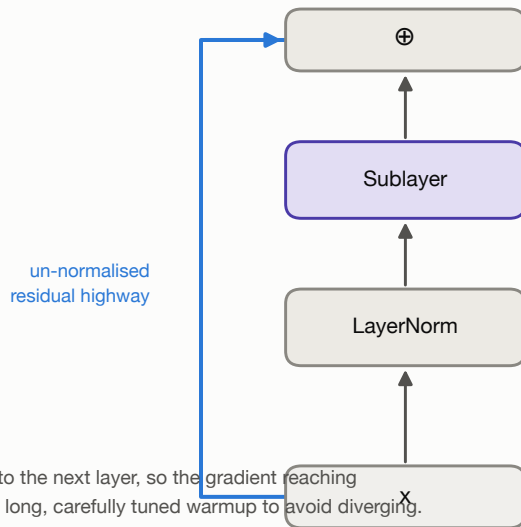


In post-LN every residual addition passes through a normalisation on its way to the next layer, so the gradient reaching layer 1 is attenuated once per layer above it — which is why post-LN needs a long, carefully tuned warmup to avoid diverging.

In pre-LN the residual path runs from input to output untouched, gradients arrive undamped, and training is stable at higher learning rates with a shorter warmup. That is why this project uses it: a Colab run that diverges at step 300 is an hour lost.

Pre-LN (used here)

$$x \leftarrow x + \text{Sublayer}(\text{LayerNorm}(x))$$



un-normalised
residual highway